

Data science Courses

1st Term:

GIU- Cairo.ICS.DS5	Machine Learning	0.7	A+	4
GIU- Cairo.ICS.DS5	Database Programming	1	A	4
GIU- Cairo.ICS.DS5	Data Engineering and Visualization	0.7	A+	4

Courses Overview — First Term

1-Database Programming

In this course, we focused on advanced database concepts and query optimization techniques using **PostgreSQL**.

Key highlights:

- Implemented **two major projects** involving complex SQL queries and optimizations.
- Applied optimization methods including **B+ Tree indexing**, **hashing**, and other widely-used techniques to enhance query performance.
- Gained practical experience in designing, managing, and tuning relational databases for efficiency.

2-Machine Learning

This course introduced us to the fundamentals of machine learning algorithms and their real-world applications.

Key highlights:

- Covered core models such as **Support Vector Machines (SVM)**, **Random Forests**, **Decision Trees**, **Logistic Regression**, and other foundational regression techniques.
 - Explored an introduction to **Deep Learning** and **Neural Networks**.
 - Completed **three competitive projects** hosted on **Kaggle**, where we worked on real datasets for **regression** and **classification** tasks to achieve the best possible predictive scores.
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3-Data Visualization

In this course, we learned effective methods for visualizing and cleaning data to extract meaningful insights.

Key highlights:

- Focused on handling **missing values**, **noise**, and **anomalies** in datasets.
- Explored different types of **plots** and visual representation methods, understanding when and how to use each.
- Completed **two practical projects** applying data preprocessing and visualization techniques to real-world datasets.

2nd Term :

Semester	Course Name	Numeric Grade	Grade	Hours
GIU- Cairo.ICS.DS6	Big Data and NoSQL	0.7	A+	4
GIU- Cairo.ICS.DS6	Advanced Machine Learning	0.7	A+	4
GIU- Cairo.ICS.DS6	Introduction to Image Processing and Computer Vision	0.7	A+	4

Courses Overview — Second Term

1- Big Data

This course focused on working with modern big data frameworks and tools.

Key highlights:

- Explored technologies such as **MongoDB**, **Apache Cassandra**, **SparkSQL**, **Spark DataFrames**, and integrated usage with **NumPy** for data operations.
 - Completed **three projects** where we designed and executed queries using these frameworks to understand their structures, query capabilities, and performance considerations.
 - Gained practical experience in handling large-scale data storage and distributed processing.
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2-Advanced Machine Learning

This course built on foundational ML knowledge and dove deeper into advanced techniques and neural networks.

Key highlights:

- Covered advanced topics starting from **Random Forests** to **Deep Learning** fundamentals.
 - Studied **Artificial Neural Networks (ANN)**, **Convolutional Neural Networks (CNN)**, and **Recurrent Neural Networks (RNN)** — including the **Sequential Model** and how outputs are passed through layers for complex tasks.
 - Explored modern concepts such as distributed learning frameworks and the underlying principles behind large language models like **ChatGPT**.
 - Completed **three projects**, competing on **Kaggle** with deep learning models (CNNs, RNNs, ANNs) for various real-world tasks.
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3-Computer Vision

This course introduced the fundamentals of computer vision and its practical applications.

Key highlights:

- Learned the principles behind visual data processing and image-based recognition systems.
- Completed **two projects**, including a facial expression detector (e.g., classifying **happy** or **sad** faces) and a **street sign detection** project, applying image processing and basic object recognition techniques.
- Built a strong understanding of how computer vision is used in modern AI applications